

JNTU Kakinada

MID-TERM EXAMINATION ? 2026

Engineering Physics ? Mid-Term Examination, 2026

Subject Code: PH102 Semester: 1 Regulation: R23

Time: 90 Minutes Max. Marks: 30

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Interference of Light)

1. Define Interference of Light. Explain its significance in Engineering Physics with a suitable example. [12 Marks]
2. Explain Interference of Light with a neat diagram/sketch and a suitable example. [12 Marks]
3. Describe the working principle of Interference of Light with necessary equations. [12 Marks]

SECTION B (Unit 2: Polarization)

4. Explain Polarization with a neat diagram/sketch and a suitable example. [12 Marks]
5. Discuss the engineering applications of Polarization. [8 Marks]
6. Discuss the engineering applications of Polarization. [12 Marks]

SECTION C (Unit 3: Quantum Mechanics)

7. Write short notes on Quantum Mechanics. [12 Marks]
8. Discuss the engineering applications of Quantum Mechanics. [12 Marks]
9. List the advantages and limitations of Quantum Mechanics. Explain any two in detail. [12 Marks]

SECTION D (Unit 4: Lasers)

10. Derive the expression related to Lasers and explain each term. [12 Marks]
11. Write short notes on Lasers. [12 Marks]
12. Write short notes on Lasers. [12 Marks]

SECTION E (Unit 5: Optical Fibers)

13. Solve a problem related to Optical Fibers. Show all the steps clearly. [12 Marks]
14. Write short notes on Optical Fibers. [8 Marks]
15. Derive the expression related to Optical Fibers and explain each term. [8 Marks]

*** End of Question Paper ***